

	be increased to compensate for the decrease in frictional force.
7(a)(i)	Background radiation is a measure of the level of ionising radiation present in the environment at a particular location which is not due to deliberate introduction of radioactive or radiation sources. Sources of background radiation:

	1. Cosmic radiation (due to radiation from outer space, from the Sun and stars) 2. Terrestrial radiation (due to decay of radioactive materials in the earth itself)
7(a)(ii)	The unstable nucleus undergoes a random disintegration in radioactive decay, in which it is impossible to determine <u>exactly which nucleus</u> and <u>exactly when</u> a particular nucleus will disintegrate.
7(b)	Small amounts of ionising radiation can damage living cells, as in the case of medical imaging procedure, but the cells can normally repair themselves. However, prolonged exposure to small amounts of ionising radiation can damage the cells beyond repair. Large amounts of ionising radiation will kills living cells and cause them to lose their abilities to repair and reproduce. Large amounts can kill so many cells that tissues and organs are damaged. Key terms: <i>effects, different amounts</i> – small and large, <i>living cells</i> [Describe requires candidates to state in words (using diagrams where appropriate) the mains points of the particular phenomena.]
7(c)(i)	Isotopes are nuclides with the <u>same number of protons</u> but <u>different number of neutrons</u> . They have the <u>same atomic number</u> but <u>different mass number</u> .
7(c)(ii)	Balanced nuclear equation for decay ${}_{95}^{241}\text{Am} \rightarrow {}_{93}^{237}\text{Np} + {}_2^4\text{He}$
7(c)(iii)	The alpha particles emitted by the americium-241 nuclei have a very short range as they loses energy rapidly due to their ionising ability. As long as the smoke detector is installed high up on a wall or the ceiling, the americium-241 nuclei are unlikely to cause harm to the occupants of the building.
7(c)(iv)	Alpha particles emitted from the nuclear decay of americium ionises air molecules, giving rise to positively and negatively charged particles and causing current to flow between two electrodes in the smoke detector. This triggers the detector to sound an alarm.
7(d)	Using $A = A_0 \exp(-\lambda t)$, $\frac{A}{A_0} = \exp(-\lambda t) \text{ where } \lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$ $0.10 = \exp\left(-\frac{\ln 2}{432} t\right)$ $\therefore t = 1435$ $\approx 1440 \text{ years}$ Alternatively,

	$\frac{A}{A_0} = \left(\frac{1}{2}\right)^n$ $0.10 = \left(\frac{1}{2}\right)^n$ $n = 3.322 \text{ half-lives}$ $\therefore t = 3.322 \times 432$ $= 1435$ $\approx 1440 \text{ years}$
7(e)(i)	Using $\Delta E = \Delta mc^2$, Overall loss in mass of an americium-241 nucleus, $\Delta m = \frac{\Delta E}{c^2}$ $= \frac{(5.49 \times 10^6)(1.60 \times 10^{-19})5.49 \times 10^6 \times 1.60}{(3.00 \times 10^8)^2}$ $= 9.76 \times 10^{-30} \text{ kg}$
7(e)(ii)	With the initial rate of 1.15×10^{12} decays per year, the initial number of americium-241 nuclide decaying is 1.15×10^{12} per year. Initial rate of loss of mass due to these decays $= (1.15 \times 10^{12})(9.76 \times 10^{-30})$ $= 1.122 \times 10^{-17}$ $\approx 1.12 \times 10^{-17} \text{ kg year}^{-1}$